# IMAGE-IDENTIFIABLE GENOMIC SUBSPACES: A LINEAR–GAUSSIAN MODEL OF RADIOGENOMIC RECOVERABILITY

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**ABSTRACT**

We develop a linear–Gaussian generative model linking patient-level genomic profiles $g \in \mathbb{R}^p$ to imaging phenotypes $x \in \mathbb{R}^d$, and use it to ask a sharper question than "does imaging predict gene $j$?" Instead we ask *which low-dimensional genomic subspaces are image-identifiable*. Under the model every quantity of interest is available in closed form: the Bayes-optimal estimator $\mathbb{E}[g \mid x]$, the posterior covariance $\Sigma_{G|X}$, the mutual information $I(g; x)$, and a per-direction *recoverability score* $R(v) \in [0, 1]$. We show that the directions of maximal recoverability solve a generalized eigenproblem that coincides exactly with canonical correlation analysis (CCA) between the two modalities, and that the recoverability of the $i$-th canonical genomic direction equals the squared canonical correlation $\rho_i^2$. We then address estimation in the regime $p \gg n$ via regularized covariances, reduced-rank regression, and the probabilistic-CCA latent-variable form, and describe cross-validated and permutation-based procedures for honest inference. The companion notebook `notebooks/radiogenomic_recoverability.ipynb` implements the full pipeline on a pair of `AnnData` objects, with a synthetic-data fallback that has known ground truth.

***Keywords*** radiogenomics · canonical correlation analysis · mutual information · recoverability · linear–Gaussian model

## 1 Setup and notation

We observe $n$ patients. For patient $i$ let $g_i \in \mathbb{R}^p$ be a vector of genomic features (e.g. expression, somatic mutation burden, copy-number scores) and $x_i \in \mathbb{R}^d$ a vector of imaging phenotypes (radiomic descriptors, deep features, or structured reads) [1–3]. Stack the observations into data matrices $\mathbf{G} \in \mathbb{R}^{n\times p}$ and $\mathbf{X} \in \mathbb{R}^{n\times d}$, with row $i$ equal to $g_i^\top$ and $x_i^\top$ respectively. Random variables are written in uppercase $(G, X)$, realizations in lowercase $(g, x)$. Throughout we assume both modalities have been centered, so all means are zero; recentring is handled by the estimators of Section 10.

We use $\Sigma_G \in \mathbb{R}^{p\times p}$, $\Sigma_X \in \mathbb{R}^{d\times d}$ for the marginal covariances, $\Sigma_{GX} = \mathrm{Cov}(G, X) \in \mathbb{R}^{p\times d}$ for the cross-covariance, and $\Sigma_{XG} = \Sigma_{GX}^\top$. For a symmetric positive semidefinite $S$ we write $S^{1/2}$ for its symmetric square root and $S^{-1/2}$ for the inverse thereof.

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## 2 The linear–Gaussian generative model

The genomic state is drawn from a Gaussian prior and the imaging phenotype is a noisy linear readout of it:

$$G \sim \mathcal{N}(0, \Sigma_G), \tag{1}$$

$$X \mid G \sim \mathcal{N}(AG,\ \sigma^2 I_d), \qquad A \in \mathbb{R}^{d\times p}. \tag{2}$$

The matrix $A$ encodes how genomic variation is expressed in the image; the isotropic term $\sigma^2 I_d$ collects everything in the image that genomics does not drive — biological variation outside the measured panel, acquisition and reconstruction noise, and feature-extraction error. Equivalently,

$$X = AG + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d), \quad \varepsilon \perp G. \tag{3}$$

Equations (1)–(3) are deliberately the simplest model with a non-trivial answer. Their value is that *every* downstream quantity is closed-form, so the model functions as an exact upper bound: no estimator — linear or nonlinear — can recover genomics from imaging better than the posterior derived below (under these assumptions). This fundamental-limit framing is in the spirit of rate–distortion theory and the information bottleneck [4–8], specialized here to a tractable linear–Gaussian channel. Section 10 relaxes the isotropic-noise and known-parameter assumptions for practical fitting.

## 3 Marginal and joint laws

**Proposition 1** (Joint Gaussian law)**.** *Under* (1)–(3)*,*

$$\begin{pmatrix} G \\ X \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A\Sigma_G & A\Sigma_G A^\top + \sigma^2 I_d \end{pmatrix}\right). \tag{4}$$

*In particular,*

$$X \sim \mathcal{N}(0, \Sigma_X), \qquad \Sigma_X = A\Sigma_G A^\top + \sigma^2 I_d, \tag{5}$$

*and the cross-covariance is*

$$\Sigma_{GX} = \mathrm{Cov}(G, X) = \Sigma_G A^\top. \tag{6}$$

*Proof.* Since

$$X = AG + \varepsilon,$$

where

$$G \sim \mathcal{N}(0, \Sigma_G), \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d),$$

and $G$ and $\varepsilon$ are independent, the stacked vector

$$\begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} G \\ AG + \varepsilon \end{pmatrix}$$

is jointly Gaussian, as it is an affine transformation of independent Gaussian variables.

It therefore suffices to compute its mean and covariance. Since both $G$ and $\varepsilon$ are centered,

$$\mathbb{E}[G] = 0, \qquad \mathbb{E}[\varepsilon] = 0,$$

we obtain

$$\mathbb{E}[X] = A\mathbb{E}[G] + \mathbb{E}[\varepsilon] = 0,$$

and hence

$$\mathbb{E}\begin{pmatrix} G \\ X \end{pmatrix} = 0.$$

Next, the covariance matrix has block form

$$\mathrm{Cov}\begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} \mathrm{Cov}(G,G) & \mathrm{Cov}(G,X) \\ \mathrm{Cov}(X,G) & \mathrm{Cov}(X,X) \end{pmatrix}.$$

The upper-left block is simply

$$\mathrm{Cov}(G, G) = \Sigma_G.$$

For the upper-right block, using $X = AG + \varepsilon$,

$$\mathrm{Cov}(G, X) = \mathrm{Cov}(G, AG + \varepsilon).$$

By linearity of covariance,

$$\mathrm{Cov}(G, X) = \mathrm{Cov}(G, AG) + \mathrm{Cov}(G, \varepsilon).$$

Since $G$ and $\varepsilon$ are independent,

$$\mathrm{Cov}(G, \varepsilon) = 0,$$

and therefore

$$\mathrm{Cov}(G, X) = \mathrm{Cov}(G, AG).$$

Using the identity

$$\mathrm{Cov}(U, BV) = \mathrm{Cov}(U, V)B^\top,$$

we obtain

$$\mathrm{Cov}(G, AG) = \Sigma_G A^\top.$$

Hence

$$\Sigma_{GX} = \mathrm{Cov}(G, X) = \Sigma_G A^\top.$$

The lower-left block is the transpose:

$$\mathrm{Cov}(X, G) = \mathrm{Cov}(G, X)^\top = A\Sigma_G.$$

Finally, for the lower-right block,

$$\mathrm{Cov}(X, X) = \mathrm{Cov}(AG + \varepsilon).$$

Expanding the covariance,

$$\mathrm{Cov}(X, X) = \mathrm{Cov}(AG) + \mathrm{Cov}(\varepsilon),$$

since the cross terms vanish by independence. Using

$$\mathrm{Cov}(AG) = A\Sigma_G A^\top,$$

together with

$$\mathrm{Cov}(\varepsilon) = \sigma^2 I_d,$$

gives

$$\mathrm{Cov}(X, X) = A\Sigma_G A^\top + \sigma^2 I_d.$$

Therefore

$$\Sigma_X = A\Sigma_G A^\top + \sigma^2 I_d.$$

Collecting the covariance blocks yields

$$\mathrm{Cov}\begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A\Sigma_G & A\Sigma_G A^\top + \sigma^2 I_d \end{pmatrix},$$

which proves the claim. $\square$

Equation (5) is the model's first interpretive statement: imaging covariance decomposes additively into a *genomics-driven* component $A\Sigma_G A^\top$ and an *irreducible* component $\sigma^2 I_d$. The cross-covariance $\Sigma_{GX} = \Sigma_G A^\top$ is the central estimable object — it names which genomic directions co-vary with imaging at all.

## 4 Bayes-optimal recovery of genomics from imaging

**Proposition 2** (Posterior)**.** *The posterior of genomics given imaging is Gaussian, $G \mid X \sim \mathcal{N}(\hat{g}(X), \Sigma_{G|X})$, with*

$$\hat{g}(X) = \mathbb{E}[G \mid X] = \Sigma_G A^\top \big(A\Sigma_G A^\top + \sigma^2 I_d\big)^{-1} X \;=\; BX, \tag{7}$$

$$\Sigma_{G|X} = \Sigma_G - \Sigma_G A^\top \big(A\Sigma_G A^\top + \sigma^2 I_d\big)^{-1} A\Sigma_G. \tag{8}$$

*Equivalently, in precision form,*

$$\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A, \qquad B = \sigma^{-2}\, \Sigma_{G|X}\, A^\top. \tag{9}$$

*Proof.* Equations (7)–(8) are the standard Gaussian conditioning formulas applied to Proposition 1. The precision form follows from the Woodbury identity: $\left(\Sigma_G^{-1} + \sigma^{-2}A^\top A\right)^{-1} = \Sigma_G - \Sigma_G A^\top(A\Sigma_G A^\top + \sigma^2 I_d)^{-1}A\Sigma_G$, and then $B = \Sigma_G A^\top \Sigma_X^{-1} = \sigma^{-2}\Sigma_{G|X}A^\top$ by the push-through identity $\Sigma_G A^\top(A\Sigma_G A^\top + \sigma^2 I)^{-1} = (\Sigma_G^{-1} + \sigma^{-2}A^\top A)^{-1}\sigma^{-2}A^\top$. □

The estimator $\hat{g}(X) = BX$ is the Bayes-optimal predictor under squared-error loss; because the model is jointly Gaussian it is also the optimal predictor over *all* measurable functions of $X$, not merely linear ones. Thus $\Sigma_{G|X}$ is a hard floor on residual genomic uncertainty: **no algorithm can do better**. The precision form (9) makes the accounting transparent — imaging *adds* information $\sigma^{-2}A^\top A$ to the prior precision $\Sigma_G^{-1}$.

## 5 Mutual information and the SNR decomposition

**Proposition 3** (Mutual information)**.** *The mutual information between the two modalities is*

$$I(G;X) = \tfrac{1}{2}\log\frac{\det\Sigma_G}{\det\Sigma_{G|X}} = \tfrac{1}{2}\log\det\left(I_d + \sigma^{-2}A\Sigma_G A^\top\right). \tag{10}$$

*Let $\lambda_1 \geq \cdots \geq \lambda_r > 0$ be the nonzero eigenvalues of $A\Sigma_G A^\top$. Then*

$$I(G;X) = \tfrac{1}{2}\sum_{i=1}^{r}\log\left(1 + \frac{\lambda_i}{\sigma^2}\right). \tag{11}$$

*Proof.* For jointly Gaussian $(G, X)$, $I(G;X) = \frac{1}{2}\log\left(\det\Sigma_G/\det\Sigma_{G|X}\right)$ [9]. Substituting (9), $\det\Sigma_{G|X} = \det\Sigma_G\,/\det(I_p + \sigma^{-2}\Sigma_G^{1/2}A^\top A\Sigma_G^{1/2})$, and Sylvester's determinant identity moves $A$ to the other side to give $\det(I_d + \sigma^{-2}A\Sigma_G A^\top)$. Diagonalizing $A\Sigma_G A^\top$ yields (11). □

Equation (11) is the cleanest reading of the model: each image-visible genomic direction contributes $\frac{1}{2}\log(1 + \mathrm{SNR}_i)$ bits (nats), where $\mathrm{SNR}_i = \lambda_i/\sigma^2$ is the signal-to-noise ratio of that direction. Directions whose genomic signal is small relative to imaging noise contribute essentially nothing, regardless of how many genes load on them. Figure 1 illustrates the decomposition (11) on the synthetic instance introduced below: the per-direction bit contribution falls off rapidly after the first handful of canonical components, and the cumulative total saturates at $I(G;X) \approx 6.45$ bits out of the analytic upper bound implied by the model.

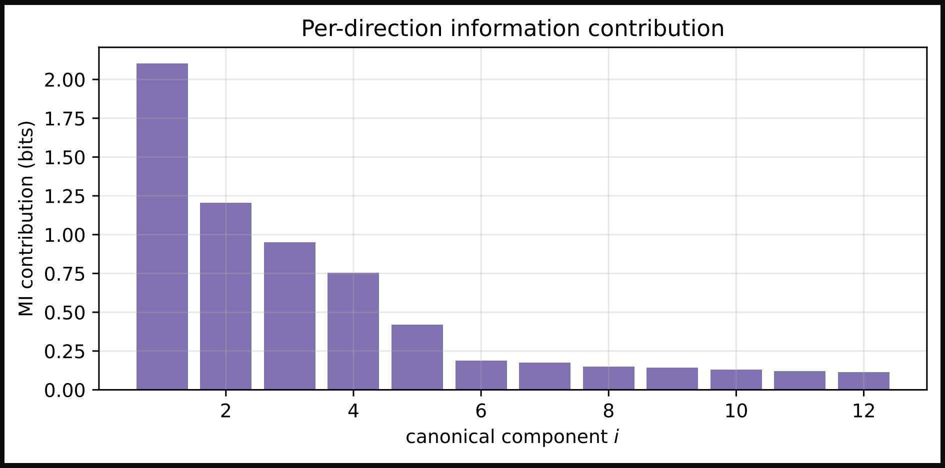


Figure 1: Per-direction contribution to $I(G;X)$ on a synthetic linear–Gaussian instance ($n = 800$, $p = 80$, $d = 40$). Each bar is $\frac{1}{2}\log(1 + d_i^2)$ in bits; the dashed line is the cumulative total.

## 6 Canonical coordinates and the recoverability spectrum

The eigenvalues $\lambda_i$ are most interpretable after whitening. Define the whitened genomic variable $U = \Sigma_G^{-1/2} G \sim \mathcal{N}(0, I_p)$ and the *whitened transfer operator*

$$M := \sigma^{-1} A\, \Sigma_G^{1/2} \in \mathbb{R}^{d\times p}, \qquad M = \sum_i d_i\, u_i v_i^\top \quad \text{(SVD)}, \tag{12}$$

with left/right singular vectors $u_i \in \mathbb{R}^d$, $v_i \in \mathbb{R}^p$ and singular values $d_1 \geq d_2 \geq \cdots \geq 0$.

**Proposition 4** (Canonical decomposition of recoverability)**.** *With $M$ as in* (12)*, define the genomic canonical direction $a_i := \Sigma_G^{-1/2} v_i$ and the canonical score $S_i := a_i^\top G = v_i^\top U$. Then:*

*(i) $S_i$ has prior variance 1 and posterior variance $\mathrm{Var}(S_i \mid X) = (1 + d_i^2)^{-1}$;*

*(ii) the canonical correlation between $S_i$ and imaging is $\rho_i = d_i/\sqrt{1+d_i^2}$, equivalently $d_i^2 = \rho_i^2/(1-\rho_i^2)$;*

*(iii) the eigenvalues of Section 5 satisfy $\lambda_i/\sigma^2 = d_i^2$, so*

$$I(G;X) = \tfrac{1}{2}\sum_i \log(1+d_i^2) = -\tfrac{1}{2}\sum_i \log(1-\rho_i^2). \tag{13}$$

*Proof.* In whitened coordinates the likelihood is $X \mid U \sim \mathcal{N}(\sigma MU, \sigma^2 I_d)$, since $AG = A\Sigma_G^{1/2} U = \sigma MU$. The posterior precision of $U$ is $I_p + \sigma^{-2}(\sigma M)^\top(\sigma M) = I_p + M^\top M = V(I_p + D^2)V^\top$, so $\mathrm{Var}(U \mid X) = V(I_p + D^2)^{-1}V^\top$ and $\mathrm{Var}(S_i \mid X) = v_i^\top \mathrm{Var}(U \mid X) v_i = (1+d_i^2)^{-1}$, giving (i). For (ii), $\mathrm{Cov}(U, X) = \mathbb{E}[U(\sigma MU + \varepsilon)^\top] = \sigma M^\top$, so $\mathrm{Cov}(S_i, X) = \sigma v_i^\top M^\top = \sigma d_i u_i^\top$. The matched imaging variate $T_i := u_i^\top X$ has $\mathrm{Var}(T_i) = u_i^\top \Sigma_X u_i = \sigma^2(1+d_i^2)$ (using $\Sigma_X = \sigma^2(MM^\top + I_d)$) and $\mathrm{Cov}(S_i, T_i) = \sigma d_i$, hence $\rho_i = \mathrm{corr}(S_i, T_i) = d_i/\sqrt{1+d_i^2}$. Part (iii) is immediate from $A\Sigma_G A^\top = \sigma^2 MM^\top$ and $\det(I_d + MM^\top) = \prod_i (1+d_i^2)$. $\square$

Part (i) gives the headline identity. Define the recoverability of a genomic score as one minus the fraction of its variance that survives observing imaging (formalized in Section 7); then

$$\boxed{R(a_i) = 1 - \mathrm{Var}(S_i \mid X) = 1 - \frac{1}{1+d_i^2} = \frac{d_i^2}{1+d_i^2} = \rho_i^2.} \tag{14}$$

**The recoverability of the $i$-th canonical genomic direction is exactly the squared canonical correlation.** This is what reduces the whole analysis to CCA between $\mathbf{G}$ and $\mathbf{X}$ (Section 10), and it means $A$ and $\sigma^2$ never have to be estimated individually — only the three second moments $\Sigma_G, \Sigma_X, \Sigma_{GX}$ enter.

## 7 Recoverability of a named genomic direction

In practice one often cares about a specific, biologically named genomic score $Y = v^\top G$ for a fixed $v \in \mathbb{R}^p$ — a pathway signature, a polygenic score, a single gene's expression ($v = e_j$).

**Definition 1** (Recoverability score)**.** *For $v \in \mathbb{R}^p$ with $v^\top \Sigma_G v > 0$,*

$$R(v) := 1 - \frac{v^\top \Sigma_{G|X}\, v}{v^\top \Sigma_G\, v} = \frac{v^\top(\Sigma_G - \Sigma_{G|X})\, v}{v^\top \Sigma_G\, v} \;\in\; [0,1]. \tag{15}$$

$R(v) = 0$ means imaging carries no information about $Y = v^\top G$; $R(v) = 1$ means imaging determines it exactly. The score is scale-invariant in $v$ and, by the Gaussian conditional-variance identity, equals the population $R^2$ of the Bayes-optimal predictor of $Y$ from $X$: $R(v) = 1 - \mathbb{E}[\mathrm{Var}(Y \mid X)]/\mathrm{Var}(Y)$. The matrix at the core of (15) has the convenient moment-only form

$$\Sigma_G - \Sigma_{G|X} = \Sigma_G A^\top \Sigma_X^{-1} A \Sigma_G = \Sigma_{GX}\, \Sigma_X^{-1}\, \Sigma_{XG}. \tag{16}$$

## 8 The recoverability eigenproblem

The natural project-level question — *which* genomic directions does imaging see best — is the variational problem

$$\max_{v\neq 0} R(v) = \max_{v\neq 0} \frac{v^\top\left(\Sigma_{GX}\Sigma_X^{-1}\Sigma_{XG}\right)v}{v^\top\Sigma_G\, v}. \tag{17}$$

**Proposition 5** (Recoverability eigenproblem = CCA)**.** *The stationary points of* (17) *are the solutions of the generalized eigenproblem*

$$\Sigma_{GX}\,\Sigma_X^{-1}\,\Sigma_{XG}\, v = \mu\,\Sigma_G\, v. \tag{18}$$

*Its eigenvalues are $\mu_i = \rho_i^2 = d_i^2/(1+d_i^2)$ and its eigenvectors are the canonical genomic directions $a_i$ of Proposition 4, $\Sigma_G$-orthonormal ($a_i^\top\Sigma_G a_j = \delta_{ij}$). Thus the ordered solutions of* (18) *are precisely the canonical correlation directions between $G$ and $X$, and $\max_v R(v) = \rho_1^2$.*

*Proof.* (18) is the first-order condition of the Rayleigh quotient (17). Substitute $v = \Sigma_G^{-1/2}w$ and $\Sigma_{GX} = \Sigma_G A^\top$; the generalized eigenproblem becomes the ordinary one $\Sigma_G^{1/2}A^\top\Sigma_X^{-1}A\Sigma_G^{1/2}\, w = \mu\, w$. Now $A\Sigma_G^{1/2} = \sigma M$ and $\Sigma_X = \sigma^2(MM^\top + I_d)$, so

$$\Sigma_G^{1/2}A^\top\Sigma_X^{-1}A\Sigma_G^{1/2} = \sigma M^\top\cdot\sigma^{-2}(MM^\top + I_d)^{-1}\cdot\sigma M = M^\top(MM^\top + I_d)^{-1}M.$$

With the SVD $M = UDV^\top$ this equals $V\,D^2(D^2+I)^{-1}\,V^\top$, whose eigenvalues are $d_i^2/(1+d_i^2) = \rho_i^2$ with eigenvectors $v_i$. Hence $w = v_i$ and $v = \Sigma_G^{-1/2}v_i = a_i$; the $\Sigma_G$-orthonormality $a_i^\top\Sigma_G a_j = v_i^\top v_j = \delta_{ij}$ is immediate. $\square$

So the deliverable of the analysis is a *recoverability spectrum* $\rho_1^2 \geq \rho_2^2 \geq \cdots$ together with the genomic directions $a_i$ that achieve it. Reading the loadings of $a_1, a_2, \ldots$ back onto genes or pathways answers "which biology is image-identifiable": the leading directions typically align with coarse, spatially expressed programs — proliferation, hypoxia, angiogenesis, necrosis, immune infiltration, tumor burden, stromal remodeling — rather than with individual transcripts, consistent with clear-cell RCC radiogenomic findings [10–12].

**Illustration on synthetic data.** Figure 2 shows the recoverability spectrum on a synthetic radiogenomic instance generated from the linear–Gaussian model of Section 2 ($n = 800$, $p = 80$, $d = 40$, latent rank $k = 5$ with prescribed $\{\rho_i^\star\}$). The estimated $\hat\rho_i^2$ closely match the planted values for the identifiable directions and decay toward the high-dimensional bulk thereafter, validating Proposition 5. Figure 3 reads the same estimator against the analytic ground truth derived from Propositions 2–4: the recovered posterior mean $\hat{B}X$, posterior-variance retention, and direction alignment all track the closed-form predictions.

## 9 Rank limits and identifiability

**Proposition 6** (Rank limit)**.** *The recoverable signal matrix has rank*

$$\operatorname{rank}(\Sigma_G - \Sigma_{G|X}) = \operatorname{rank}(A\Sigma_G A^\top) \leq \min\bigl(d,\ p,\ \operatorname{rank} A,\ \operatorname{rank}\Sigma_G\bigr). \tag{19}$$

*Consequently at most $\min(d, p, \operatorname{rank} A, \operatorname{rank}\Sigma_G)$ canonical correlations are nonzero, and the image-identifiable genomic subspace $\operatorname{span}\{a_i : \rho_i > 0\}$ has dimension bounded by the same quantity.*

This is the conceptual core. Even with $p = 20{,}000$ measured genes, imaging can expose only a projection of genomics whose dimension is capped by the rank of the transfer map $A$ — if a CT phenotype reflects, say, 20 macroscopic biological axes, then no model recovers 20,000 independent molecular variables from it. The recoverable dimension is a property of the *biology–imaging channel*, not of the estimator. A data-processing remark: imaging features are themselves a deterministic, lossy function $x = \phi(\text{image})$; any such post-processing can only shrink $\operatorname{rank}(\Sigma_{GX})$ and lower each $\rho_i$, never raise them, by the data processing inequality [7, 9, 13–15].

## 10 Estimation from finite samples

In applications $p$ and $d$ may rival or exceed $n$, the noise is not isotropic, and $A, \Sigma_G, \sigma^2$ are unknown. Because Propositions 4 and 5 require only $\Sigma_G, \Sigma_X, \Sigma_{GX}$, we estimate those directly.

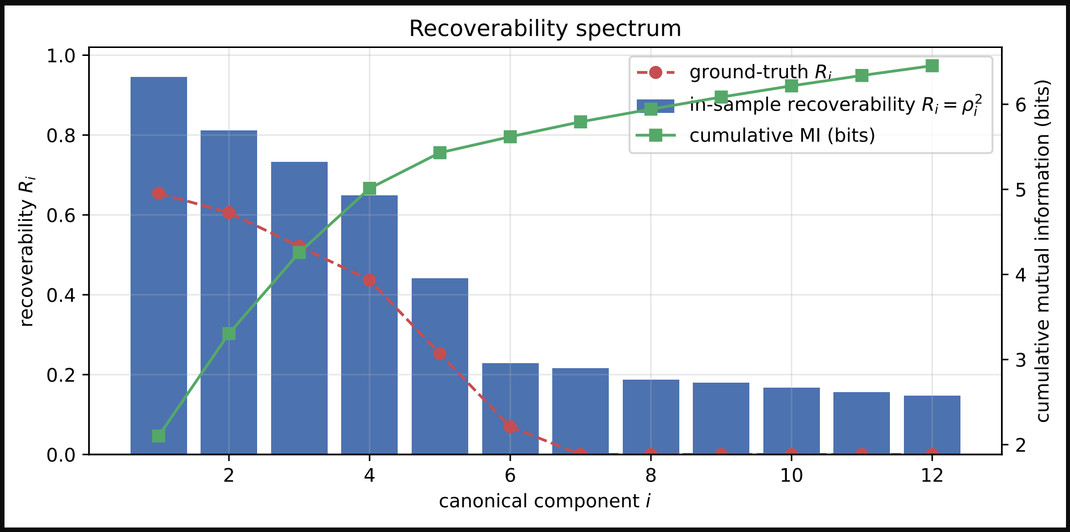


Figure 2: Recoverability spectrum on a synthetic linear–Gaussian instance ($n = 800$, $p = 80$, $d = 40$). The leading $\hat{\rho}_i^2$ recovers the planted canonical structure; later components fall to the high-dimensional bulk.

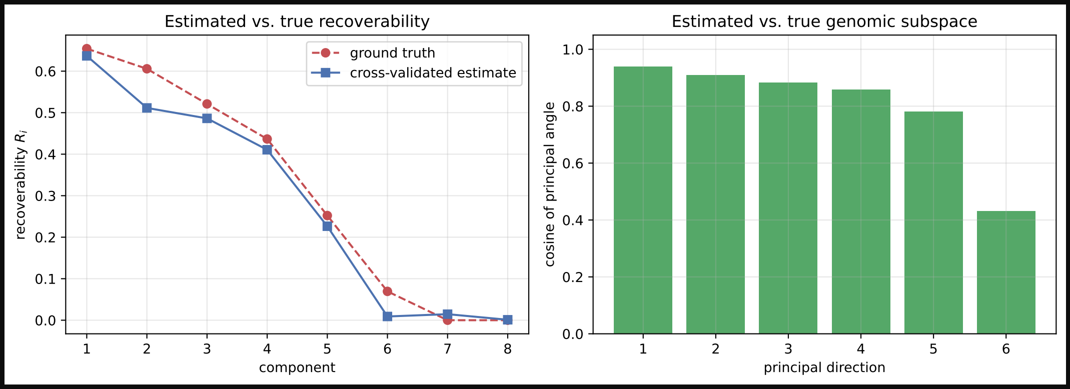


Figure 3: Empirical-vs-analytic check on the same synthetic instance. Estimated canonical correlations, recoverability per direction, and imaging variance attributable to genomics all match the closed-form quantities from Propositions 2–4.

### 10.1 Regularized covariances

Replace sample covariances by shrinkage estimators (e.g. Ledoit–Wolf [16]), $\hat{\Sigma_G} = (1 - \alpha_G)\, S_G + \alpha_G\, \nu_G I_p$ and likewise $\hat{\Sigma_X}$, with $S_G = n^{-1}\mathbf{G}^\top\mathbf{G}$ the centered sample covariance; the cross-covariance is $\hat{\Sigma_{GX}} = n^{-1}\mathbf{G}^\top\mathbf{X}$. Shrinkage keeps $\hat{\Sigma_G}, \hat{\Sigma_X}$ invertible and well-conditioned, which is required for the whitening in (12). When $p$ is very large it is practical to first project genomics onto its top principal components (or onto curated pathway scores), reducing $p$ to a manageable $p' \ll n$ before estimation; the rank limit of Proposition 6 means little is lost provided $p'$ exceeds the recoverable dimension.

### 10.2 Reduced-rank and ridge regression

The transfer map, if needed explicitly, is the multivariate regression of $X$ on $G$. Ridge gives $\hat{A}^\top = (\mathbf{G}^\top\mathbf{G} + \gamma I_p)^{-1}\mathbf{G}^\top\mathbf{X}$, and $\hat{\sigma}^2$ follows from the residual covariance. Imposing $\mathrm{rank}(A) \le k$ (reduced-rank regression) bakes Proposition 6 into the fit and is equivalent, up to regularization, to truncating the SVD of $M$ at $k$ components.

### 10.3 Regularized CCA

The estimator used in the companion notebook solves (18) with the regularized covariances of Section 10.1: form $\hat{M} = \hat{\Sigma_G}^{-1/2}\hat{\Sigma_{GX}}\hat{\Sigma_X}^{-1/2}$, take its SVD $\hat{M} = \hat{U}\hat{D}\hat{V}^\top$, and read off $\hat{\rho}_i = \hat{d}_i$ (clipped to $[0, 1)$), $\hat{R}_i = \hat{\rho}_i^2$, genomic directions $\hat{a}_i = \hat{\Sigma_G}^{-1/2}\hat{u}_i$, imaging directions $\hat{b}_i = \hat{\Sigma_X}^{-1/2}\hat{v}_i$. The Bayes-optimal predictor and posterior covariance follow from (7)–(8) with hats throughout. This is exactly regularized CCA — the recoverability program and CCA are the same computation.

### 10.4 Probabilistic CCA: the latent-variable view

The rank limit motivates an explicit low-dimensional generative form. With a shared latent $Z \in \mathbb{R}^k$,

$$Z \sim \mathcal{N}(0, I_k), \qquad G = W_G Z + \mu_G + \epsilon_G, \qquad X = W_X Z + \mu_X + \epsilon_X, \tag{20}$$

$\epsilon_G \sim \mathcal{N}(0, \Psi_G)$, $\epsilon_X \sim \mathcal{N}(0, \Psi_X)$ independent (probabilistic CCA; Bach & Jordan, 2005). Here $k$ *is* the dimension of the image-identifiable genomic subspace, $Z$ is the shared biological state, and $\epsilon_G, \epsilon_X$ are modality-private variation. The implied moments are $\Sigma_G = W_G W_G^\top + \Psi_G$, $\Sigma_X = W_X W_X^\top + \Psi_X$, $\Sigma_{GX} = W_G W_X^\top$; the maximum-likelihood loadings are the canonical directions scaled by the canonical correlations, recovering Section 10.3. The model is fit by EM, which also yields posterior latent estimates $\mathbb{E}[Z \mid x]$ and naturally accommodates anisotropic (per-feature) noise, unlike the isotropic $\sigma^2 I_d$ of the base model.

### 10.5 Sample-size regimes and dimensional collapse

Equations (7)–(18) are population identities; the estimators of Sections 10.1–10.4 inherit a finite-sample bias whose magnitude is governed entirely by the aspect ratios $p/n$ and $d/n$. The bias is not benign, and below a critical sample size the recoverability spectrum breaks down in a specific, predictable way.

**Three regimes.** Let $p^\star, d^\star$ be the working dimensions after any pre-reduction (e.g. PCA), and write $\kappa_G = p^\star/n$, $\kappa_X = d^\star/n$.

1. *Classical* ($\kappa_G + \kappa_X \ll 1$). Sample CCA is consistent; $\hat{\rho}_i \to \rho_i$ at the usual $n^{-1/2}$ rate and the in-sample spectrum is a faithful estimate of the population one.
2. *High-dimensional* ($\kappa_G + \kappa_X \lesssim 1$, both bounded away from 1). Under the null $\Sigma_{GX} = 0$, the squared sample canonical correlations follow a Wachter-type law on $[(\sqrt{\kappa_G(1-\kappa_X)} - \sqrt{\kappa_X(1-\kappa_G)})^2, (\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2]$ [17, 18]; the leading sample value already sits well above zero by geometry. Estimates of $\rho_i$ are systematically inflated and ordinary CCA must be replaced by its regularized form, with shrinkage parameters tuned by cross-validation.
3. *Saturating* ($p^\star + d^\star \geq n$). The rows of $\mathbf{G}$ and $\mathbf{X}$ live in subspaces whose intersection has dimension at least $p^\star + d^\star - n + 1$; consequently $\hat{\rho}_1 = \cdots = \hat{\rho}_q = 1$ for $q = \min(p^\star, d^\star) - (n - \max(p^\star, d^\star))_+$ *regardless of any underlying association*. The empirical mutual information diverges, the in-sample $\mathrm{tr}(\hat{\Sigma_{GX}}\hat{\Sigma_X}^{-1}\hat{\Sigma_{XG}})/\,\mathrm{tr}(\hat{\Sigma_X})$ approaches the rank fraction of the genomic span, and the permutation null collapses onto the observed value (its $p$-values concentrate near 1). Nothing in the spectrum is estimable in this regime without shrinkage *plus* dimension reduction; shrinkage alone keeps the matrices invertible but does not restore identifiability.

**Quantitative warning signs.** Three diagnostics, jointly, separate real signal from dimensional artefact:

- the gap between in-sample $\hat{R}_i$ and the $K$-fold value $\hat{R}_i^{\mathrm{cv}}$; under regime 3 this gap approaches $\hat{R}_i - 0$;
- the permutation null for $\hat{\rho}_1$: if the observed value lies inside the bulk, the result is consistent with chance;
- the cross-validated $R^2$ of the leading canonical score $a_1^\top g$ predicted from $x$ on a held-out split; in regime 3 it is typically large and *negative* (worse than predicting the mean), a behaviour seen empirically on TCGA-KIRC at $n = 131$ with $p^\star = 100, d^\star = 107$ in the companion notebook.

**Recommendations.** Choose $p^\star$ and $d^\star$ *adaptively from* $n$, not from the native modality dimensions:

- Target $p^\star + d^\star \leq n/\tau$ with $\tau \geq 5$ as a working rule (we use $p^\star, d^\star \leq \lfloor n/\tau \rfloor$ per side in the notebook; $\tau \in [5, 10]$ trades bias for resolution). Reduce both modalities by PCA when needed — reducing only the larger side leaves the estimator in regime 3.

- Treat $\hat{R}_i^{\mathrm{cv}}$ (Section 10.6) as the primary estimand and report the in-sample $\hat{R}_i$ only with the gap made explicit; the in-sample fraction of imaging variance attributable to genomics should be reported alongside a cross-validated counterpart, e.g. $\sum_i \hat{R}_i^{\mathrm{cv}} / \min(p^\star, d^\star)$.
- Power planning. The leading population correlation $\rho_1$ is detectable above the Wachter bulk only when its squared value exceeds $(\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2$; solving for $n$ gives a minimum sample size given a hypothesised effect. For $\rho_1^2 = 0.2$ and a balanced $p^\star = d^\star$, this requires roughly $n \gtrsim 25\,p^\star$, an order of magnitude beyond what cohort sizes in radiogenomics usually offer — reinforcing the need for aggressive pre-reduction.
- Probabilistic CCA (Section 10.4) with a small, fixed latent dimension $k$ is the most parsimonious option when $n$ is the binding constraint; rank-$k$ regularization sidesteps the high-dimensional bulk by construction.

The practical message is conservative: in radiogenomic cohorts, sample size is almost always the binding constraint and a defensible analysis is the one whose effective $p^\star + d^\star$ is a small fraction of $n$, with both sides reduced symmetrically and every reported number a cross-validated or permutation-calibrated one.

Figure 4 illustrates these regimes on the synthetic instance, where the planted truth is known. As $n$ grows with $p^\star, d^\star$ scaled to keep $\kappa_G + \kappa_X$ bounded, the in-sample $\hat{\rho}_1$ sits at or near 1 across the small-$n$ saturating regime — indistinguishable from the permutation null — while the cross-validated $\hat{R}_1^{\mathrm{cv}}$ only separates from zero once $n$ clears the Wachter edge. The empirical mutual information shows the same threshold behaviour.

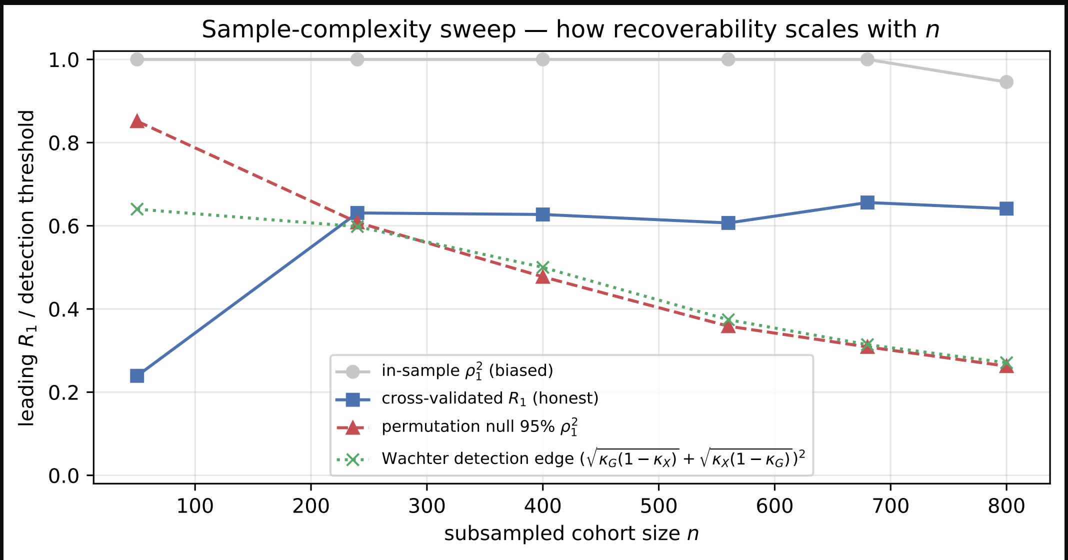


Figure 4: Sample-complexity sweep on the synthetic instance with known truth. In-sample $\hat{\rho}_1$ tracks the null until $n$ exceeds the Wachter edge; cross-validated $\hat{R}_1^{\mathrm{cv}}$ and empirical $\hat{I}(G; X)$ only become informative past that threshold. Compare to Figure 10 for the same diagnostic on TCGA-KIRC.

### 10.6 Honest recoverability: cross-validation and permutation

In-sample canonical correlations are biased upward, severely so when $p, d$ are not small relative to $n$. Two safeguards, both implemented in the notebook:

- *Cross-validation.* Estimate $\hat{a}_i, \hat{b}_i$ on a training fold; on the held-out fold form the scores $\hat{a}_i^\top g$ and $\hat{b}_i^\top x$ and report their out-of-sample squared correlation as the honest $\hat{R}_i$. The gap to the in-sample value quantifies overfitting.
- *Permutation test.* Repeatedly permute the patient labels of one modality, refit, and collect the leading canonical correlation to build a null distribution; the permutation $p$-value tests whether any genomic direction is image-identifiable beyond chance.

Figure 5 shows both diagnostics on the synthetic instance. The in-sample $\hat{R}_i$ overshoots the planted truth by the amount predicted by the high-dimensional bias, while the 5-fold $\hat{R}_i^{\mathrm{cv}}$ tracks the truth for identifiable directions and collapses toward the null 95% quantile thereafter. Figure 6 shows the permutation null on $\hat{\rho}_1$: when the true signal is nonzero, the observed value sits cleanly outside the null bulk and the per-direction $p$-values fall below 0.01 for the top three components, matching the planted rank.

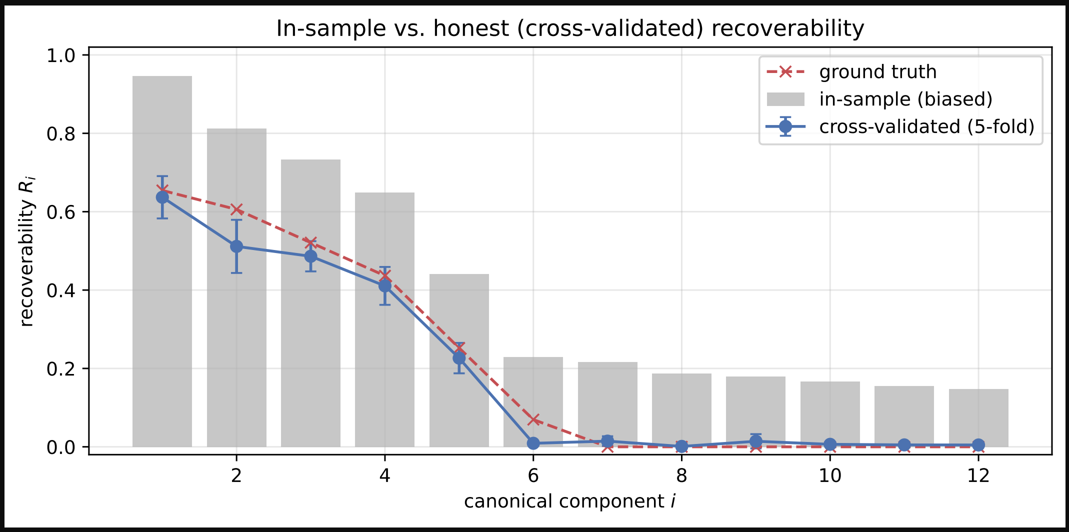


Figure 5: Cross-validated recoverability on a synthetic instance with planted rank $k = 5$. In-sample $\hat{R}_i$ (grey) is biased upward; 5-fold $\hat{R}_i^{\mathrm{cv}}$ (blue) tracks the planted truth for the identifiable directions and collapses to the null floor afterwards.

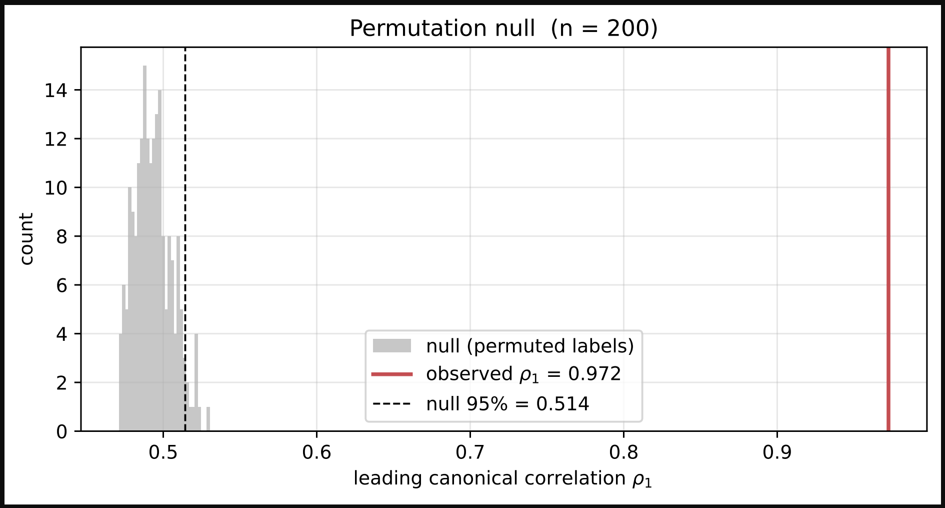


Figure 6: Permutation null on the leading canonical correlation $\hat{\rho}_1$ for the synthetic instance. Red marks the observed value; $p < 0.01$ for the top three directions, recovering the planted rank.

The same optimism appears in the aggregate fraction of imaging variance attributable to genomics. Figure 7 shows the in-sample share $\sum_i \hat{R}_i / \min(p^\star, d^\star)$ alongside its cross-validated counterpart on the synthetic instance: the in-sample number sits well above the planted truth while the CV value tracks it. The same diagnostic on TCGA-KIRC (Section 13) reports 0.193 in-sample against 0.012 cross-validated — a sixteen-fold gap that is consistent with the synthetic mechanism rather than with any genuinely recoverable signal.

## 11 Interpretation for radiogenomics

The model reframes the radiogenomic question [2, 19–21]. Rather than testing whether CT predicts each gene — a high-dimensional, low-power multiple-testing problem — it asks for the *recoverability spectrum* $\{\rho_i^2\}$ and the associated

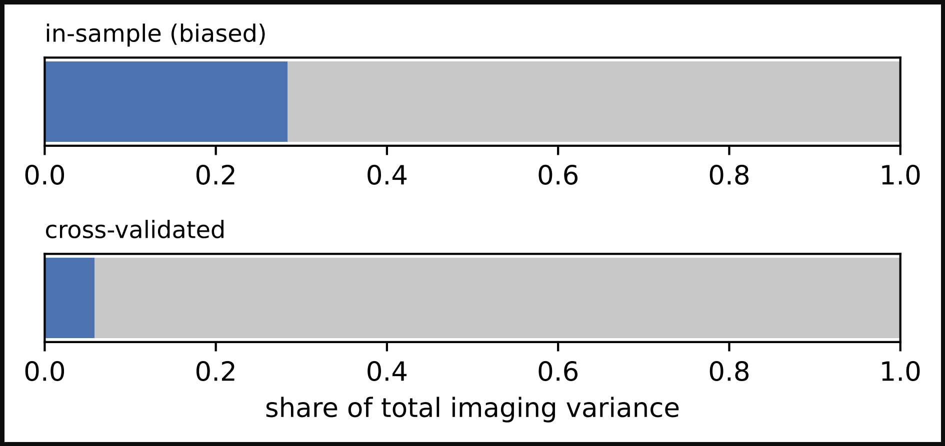


Figure 7: Share of imaging variance attributable to genomics on the synthetic instance: in-sample (grey) versus 5-fold cross-validated (blue), with the planted truth marked. The in-sample share is biased upward by the high-dimensional inflation described in Section 10.5.

genomic directions $\{a_i\}$. The outputs are: (1) a small number of image-identifiable genomic axes, with gene/pathway loadings that make them biologically legible; (2) a hard bound $\Sigma_{G|X}$ on what any predictor can achieve, per direction; (3) a total information budget $I(G;X)$; and (4) an honest, cross-validated estimate of each. The governing principle is that radiogenomics should target *image-identifiable low-dimensional genomic subspaces*, whose dimension is set by the biology–imaging channel and not by the gene panel size.

## 12 A Bernoulli extension for mutation data via a Gaussian copula

The linear–Gaussian model of Section 1 is well matched to continuous, approximately log-normal genomic readouts: log-CPM gene expression, methylation $\beta$- or $M$-values, log-CNV ratios. It is plainly misspecified for somatic mutation calls, where each $G_j \in \{0,1\}$ indicates the presence of a non-silent variant in gene $j$ and the marginal frequencies $p_j$ are small, gene-specific, and highly heterogeneous. This section extends the framework to such data by replacing the Gaussian prior on $G$ with a *Gaussian-copula Bernoulli* (equivalently, multivariate probit) prior, while preserving the canonical-correlation / mutual-information machinery of Sections 5–6.

### 12.1 The latent-Gaussian generative model

Let $p_j = \Pr(G_j = 1)$ and $\tau_j = \Phi^{-1}(1-p_j)$, where $\Phi$ is the standard normal CDF. Introduce a latent vector

$$Z \sim \mathcal{N}(0, \Sigma_Z), \qquad \operatorname{diag}(\Sigma_Z) = I_p,$$

and define the observed mutation indicator by thresholding:

$$G_j = \mathbf{1}\{Z_j > \tau_j\}, \qquad j = 1, \dots, p.$$

By construction the marginals are correct, $G_j \sim \operatorname{Bern}(p_j)$, and the pairwise joint laws are governed by the tetrachoric correlation $\Sigma_{Z,jk}$, which can be any positive-definite correlation matrix — providing arbitrary dependence structure between mutated genes (e.g. mutual exclusivity, co-occurrence within pathways).

Imaging is coupled to the *latent* state, not to the indicators directly:

$$X \mid Z \sim \mathcal{N}\big(B\,Z,\ \sigma^2 I_d\big), \qquad B \in \mathbb{R}^{d\times p}. \tag{21}$$

This is the natural specification: the biological mechanism that produces an imageable phenotype (proliferation, vascularity, necrosis) responds to an underlying continuous burden, of which an observed mutation is a coarse, binarized readout. The joint structure factors as

$$Z \to G, \qquad Z \to X,$$

so $G$ and $X$ are conditionally independent given $Z$.

### 12.2 Mutual information and a tight DPI bound

Because $G$ is a deterministic, memoryless function of $Z$, the data processing inequality yields the upper bound

$$I(G;X) \;\le\; I(Z;X), \tag{22}$$

with equality only if $Z$ can be recovered from $G$ (impossible whenever any $p_j \in (0,1)$, since thresholding is lossy). The right-hand side reduces exactly to the Gaussian formula of Section 5: if $\{\rho_i\}_{i=1}^k$ denote the canonical correlations of $(Z, X)$ under (21), then

$$I(Z;X) \;=\; -\tfrac{1}{2}\sum_{i=1}^{k}\log\bigl(1-\rho_i^2\bigr), \qquad k = \operatorname{rank}\bigl(\Sigma_Z^{1/2}B^\top\Sigma_X^{-1}B\Sigma_Z^{1/2}\bigr), \tag{23}$$

with $\Sigma_X = B\Sigma_Z B^\top + \sigma^2 I_d$. The recoverability spectrum, the directions $\{a_i\}$, and the eigenproblem of Section 8 all transfer verbatim, with $\Sigma_G$ replaced by $\Sigma_Z$.

**The binarization gap.** The slack in (22) is

$$I(Z;X) - I(G;X) \;=\; H(Z \mid G) - H(Z \mid G, X) \;=\; I(Z; X \mid G),$$

the residual information about $Z$ not already revealed by the mutation call. For a single gene with frequency $p_j$, the entropy lost to binarization is $H(Z_j) - H(G_j) = \frac{1}{2}\log(2\pi e) - h_2(p_j)$ where $h_2$ is the binary entropy in nats; rare mutations ($p_j \ll 1/2$) are highly lossy, common ones near $p_j = 1/2$ less so. Thus (22) is tightest for cohorts dominated by intermediate-frequency mutations and loosest for rare-variant panels — a useful guide to when the bound is informative.

### 12.3 Per-gene recoverability and AUC ceilings

For a named mutation $G_j$, the Bayes-optimal predictor from $X$ is the posterior probability

$$\pi_j(x) \;=\; \Pr(G_j = 1 \mid X = x) \;=\; \Phi\!\left(\frac{\mu_j(x) - \tau_j}{s_j}\right),$$

where $(\mu_j(x), s_j^2) \;=\; (\mathbb{E}[Z_j \mid X = x],\ \operatorname{Var}(Z_j \mid X = x))$ are the linear-Gaussian posterior moments for the latent coordinate, computed from $(\Sigma_Z, B, \sigma^2)$ exactly as in Section 4. The *latent recoverability* $R_j^Z = \operatorname{Var}(\mathbb{E}[Z_j \mid X])/\operatorname{Var}(Z_j)$ is the direct analogue of $R(v)$ in the Gaussian case, and the binary AUC of any predictor $\hat{G}_j(x)$ is bounded by

$$\mathrm{AUC}_j \;\le\; \Phi\!\left(\sqrt{\tfrac{R_j^Z}{1-R_j^Z}}\Big/\sqrt{2}\right), \tag{24}$$

attained by the Bayes classifier $\pi_j(x) > p_j$. Equation (24) is the natural object to compare against the empirical per-gene $\mathrm{AUC}_{\mathrm{LR}}$ and $\mathrm{AUC}_{\mathrm{RF}}$ of Section 13.6: an empirical AUC near 0.5 is consistent with either a small $R_j^Z$ *or* a degenerate $p_j$, and (24) disentangles the two.

### 12.4 Estimation

The estimator splits into three independent pieces, each well-studied:

(i) **Marginal thresholds.** Estimate $\hat{p}_j = \bar{G}_{\cdot j}$ and set $\hat{\tau}_j = \Phi^{-1}(1 - \hat{p}_j)$. Apply a continuity correction or empirical-Bayes shrinkage for rare mutations.

(ii) **Latent correlation $\hat{\Sigma}_Z$.** Estimate pairwise tetrachoric correlations from $2{\times}2$ tables of $(G_j, G_k)$ by maximum likelihood under the bivariate-normal copula, then project the resulting matrix to the nearest positive-definite correlation matrix (e.g. Higham's `nearcorr` [22]). For high-dimensional panels apply Ledoit–Wolf shrinkage as in Section 10.1.

(iii) **Cross-modal covariance $\hat{\Sigma}_{ZX}$.** The point-biserial covariance $\operatorname{Cov}(G_j, X_\ell)$ is related to the latent covariance by

$$\operatorname{Cov}(G_j, X_\ell) \;=\; \phi(\tau_j)\,\operatorname{Cov}(Z_j, X_\ell),$$

where $\phi$ is the standard normal density. Estimate $\operatorname{Cov}(G_j, X_\ell)$ from the sample and rescale by $\phi(\hat{\tau}_j)$ to obtain $\hat{\Sigma}_{ZX,j\ell}$.

With $(\hat{\Sigma}_Z, \hat{\Sigma}_{ZX}, \hat{\Sigma}_X)$ in hand, the regularized CCA of Section 10.3, the permutation null of Section 10.6, and the empirical MI bound (estimated, e.g., by $k$-nearest-neighbour or neural estimators [23–25]) carry over without further change. The dimensional-collapse warnings of Section 10.5 are if anything *more* acute, since tetrachoric estimation has higher variance than Pearson correlation, particularly for rare-rare gene pairs.

### 12.5 Mixed-modality genomics

In practice a single cohort carries both continuous (expression, CNV) and binary (mutation) features. The copula formulation accommodates this by partitioning the latent vector as $Z = (Z^{\text{cts}}, Z^{\text{bin}})$ with a single joint $\Sigma_Z$. The continuous block uses the identity link ($G^{\text{cts}} = Z^{\text{cts}}$ up to marginal standardization) and the binary block uses the probit link of Section 12.1. Estimation of $\Sigma_Z$ proceeds blockwise — Pearson within the continuous block, tetrachoric within the binary block, polyserial across blocks — after which the rest of the pipeline is identical. This is the route by which the present manuscript's gene-expression analysis (Section 13) and the per-gene mutation validation (Section 13.6) can be unified into a single coherent recoverability calculation.

## 13 Application: TCGA-KIRC

We now instantiate the framework on a real radiogenomic cohort — TCGA-KIRC (clear-cell renal cell carcinoma) [26] — to demonstrate how the recoverability spectrum, the cross-validated and permutation-calibrated diagnostics of Section 10.6, and the dimensional-collapse warnings of Section 10.5 interact on a typical cohort. The point of this section is *not* to advertise a predictor: the result is a null one. The point is to convert that null into a quantitative empirical ceiling on the imaging-to-genomics channel and a sample-size design recommendation, exactly as the synthetic experiments of Sections 6–10.6 predict. All numbers below are reproducibly generated by the companion notebook and saved to `notebooks/figures/tcga_kirc/gene_expression/tumor_radimagenet/stats.json`.

### 13.1 Data and processing pipeline

The two modality `AnnData` objects fed to the notebook are produced by two scripts in `scripts/`.

**Imaging (`process_imaging_tcga_kirc.py` → `make_imaging_matrix.py`).** For each TCGA-KIRC patient we obtain the venous-phase abdominal CT volume (`0502_VENOUS.nii`) and a manually curated tumour mask. The pipeline:

1. Pull the canonical TCIA volumes [27, 28] and the radiologist-drawn tumour segmentations; orient every volume into the canonical nibabel orientation.
2. Run TotalSegmentator [29] with the `kidney_left`/`kidney_right` target list to produce an organ mask; combine with the tumour mask to form (tumour, organ, tumour∪organ) mask triples per patient. Small blobs are removed, holes filled, and a morphological closing applied to stabilise mask boundaries.
3. Extract patient-level imaging features. We use a pretrained *RadImageNet* [30] 2D backbone applied tile-wise within the tumour mask (`rgit.radimagenet.get_radimagenet_embeddings`), producing a $d_{\text{native}} = 1841$-dimensional embedding per patient. (A second pipeline using PyRadiomics [31, 32] shape/texture features is supported for ablations.)
4. Stack patient-level embeddings into an `AnnData` object whose `obs` index is the TCGA `patient_id` and whose `var` index is the embedding-dimension labels. Output: `data/tcga_kirc/imaging/tumor_radimagenet.h5ad`.

**Genomics (`make_genomics_matrix.py`).** For each matched patient we pull the TCGA-KIRC RNA-seq counts (raw STAR counts from the GDC) and the somatic-mutation MAFs.

1. Counts are CPM-normalised, $\log_{1+}$-transformed, and reduced to the top 2000 highly variable genes (Seurat-style HVG selection in `scanpy.pp.highly_variable_genes` [33]); the resulting matrix is stored in the primary `AnnData`.
2. Somatic mutations are folded into a sparse binary gene-symbol matrix on the same patient index for downstream per-gene validation (Section 13.6).
3. Output: `data/tcga_kirc/genomics/gene_expression.h5ad`.

After intersecting patients across both `AnnData` objects we retain $n = 190$ patients with $p_{\text{native}} = 2000$ HVG transcripts and $d_{\text{native}} = 1841$ RadImageNet features.

### 13.2 Working-dimension budget

Both modalities are $z$-scored per feature and PCA-reduced symmetrically to working dimensions $p^\star = d^\star = 38 = \lfloor n/5 \rfloor$ following the $p^\star + d^\star \le n/\tau$ rule of Section 10.5 ($\tau = 5$). This places the analysis at $\kappa_G + \kappa_X = 76/190 \approx 0.4$ — the high-dimensional regime, not the saturating one, but well inside the territory where in-sample canonical correlations are heavily biased upward.

### 13.3 The recoverability spectrum and its null

Fitting regularised CCA (Ledoit–Wolf shrinkage on both sides) returns an in-sample spectrum dominated by a near-unit leading correlation ($\hat\rho_1 \approx 1.00$, $\hat R_1 \approx 1.00$) and a total in-sample mutual information of 21.3 bits over 12 retained components (Fig. 8). This is exactly the geometric artefact described in regime 2–3 of Section 10.5: 5-fold cross-validation collapses $\hat R_1$ from $\approx 1$ to $\hat R_1^{\mathrm{cv}} = 0.094$, and held-out prediction of the leading canonical genomic score has $R^2 = -3.74$ — worse than predicting the mean. The companion permutation test ($N_{\mathrm{perm}} = 200$) returns $p$-values $0.32, 0.16, 0.24$ for the top three canonical correlations. None of the leading directions are distinguishable from chance.

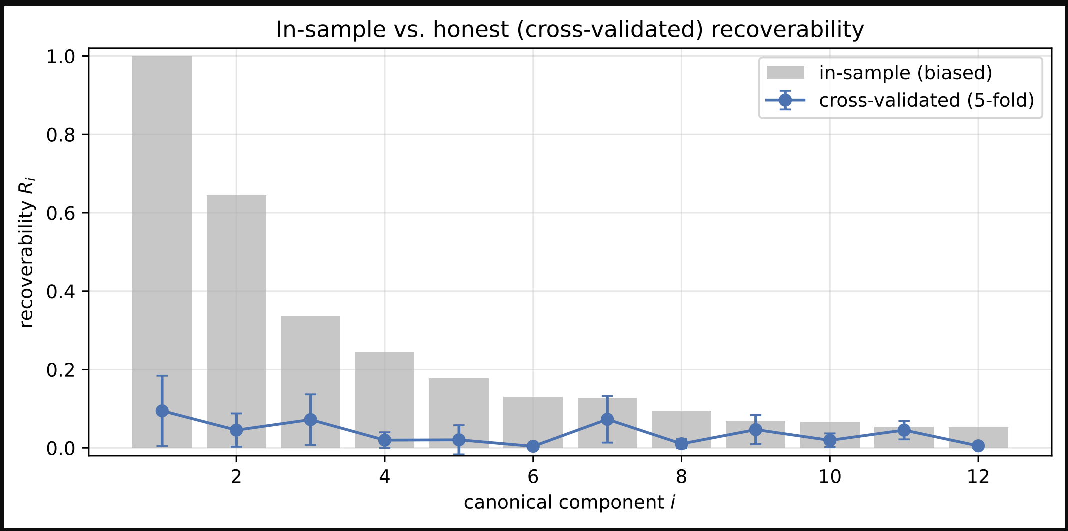


Figure 8: Recoverability spectrum on TCGA-KIRC. In-sample $\hat R_i$ (grey) tracks the high-dimensional bulk; cross-validated $\hat R_i$ (blue, 5-fold) collapses to a flat residual near zero. The gap is the overfitting predicted in Section 10.5, regime 2–3. Contrast with the synthetic instance of Fig. 5, where CV tracks the planted truth.

### 13.4 An empirical upper bound on $I(G;X)$

Section 10.5 predicts that, in this regime, the permutation null on $\hat\rho_1$ should itself be pulled toward 1 by the high-dimensional geometry; this is borne out: the per-permutation $\hat\rho_1^{\mathrm{null}}$ has mean 0.84 and a 95% quantile of 1.00. We therefore convert the test into a one-sided upper confidence limit by aggregating, across permutations, the cumulative mutual information $\mathrm{MI}^{\mathrm{null}}(K) = -\frac{1}{2}\sum_{i \le K} \log(1 - \hat\rho^2_{i,\mathrm{null}})$ over the top $K = 3$ tested components, and reading off its 95th percentile (Fig. 9):

$$I(G;X)\,\big|_{\text{KIRC, top-3}} \le 21.2 \text{ bits} \quad \text{(one-sided 95\% UCL)}, \tag{25}$$

with observed cumulative MI 20.5 bits sitting close to the null mean of 8.0 bits expressed in bits (5.5 nats). The bound is wide — this is the honest finding, not a weakness of the method — because the null is itself saturating; tightening it requires either larger $n$ or smaller $p^\star, d^\star$.

A more useful summary is the *effective image-identifiable rank*: the number of canonical components whose cross-validated $\hat R_i^{\mathrm{cv}}$ exceeds the per-component permutation-null 95% quantile. On this cohort,

$$\hat k_{\mathrm{eff}} = 0 \text{ (of } K = 3 \text{ tested)}, \tag{26}$$

so the empirical instantiation of the rank bound of Proposition 6, after sample-size deflation, is zero.

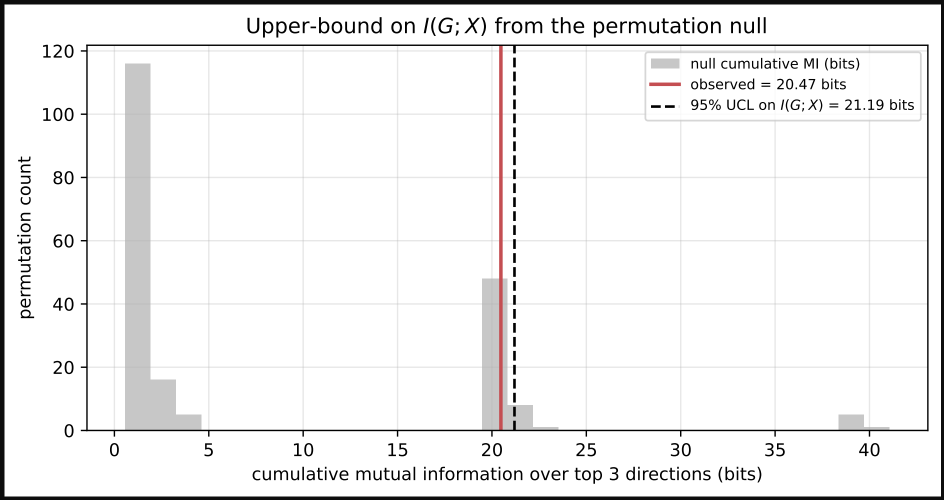


Figure 9: Cumulative-MI null distribution (top-3 canonical components) on TCGA-KIRC. The observed value (red) lies within the null bulk; the dashed line marks the $95\%$ upper confidence limit on $I(G;X)$ used in (25).

### 13.5 Sample-complexity sweep

A single-$n$ null result is most informative when paired with the scaling of that null in $n$. We subsample the cohort to $n \in \{50, 57, 95, 133, 162, 190\}$, refit the working pipeline (same $p^\star + d^\star \le n/\tau$ budget, 5-fold CV, 100 permutations) and plot the three observables vs. $n$ (Fig. 10). The cross-validated $\hat{R}_1^{\mathrm{cv}}$ is flat at $\approx 0.05$–$0.10$ across the full range; doubling $n$ does not move it. The Wachter detection edge sits at $\approx 0.63$ throughout, because $p^\star, d^\star$ grow with $n$ under our rule, so the cohort-statistical ceiling for distinguishing $\rho_1$ from the high-dimensional bulk would require $\rho_1^2 \gtrsim 0.6$ — well above anything observed at any subsample size. Inverting the Wachter expression for a target $\rho_1^2 = 0.2$ at balanced $p^\star = d^\star$ gives the sample-size requirement $n \gtrsim 25\, p^\star$ quoted in Section 10.5.

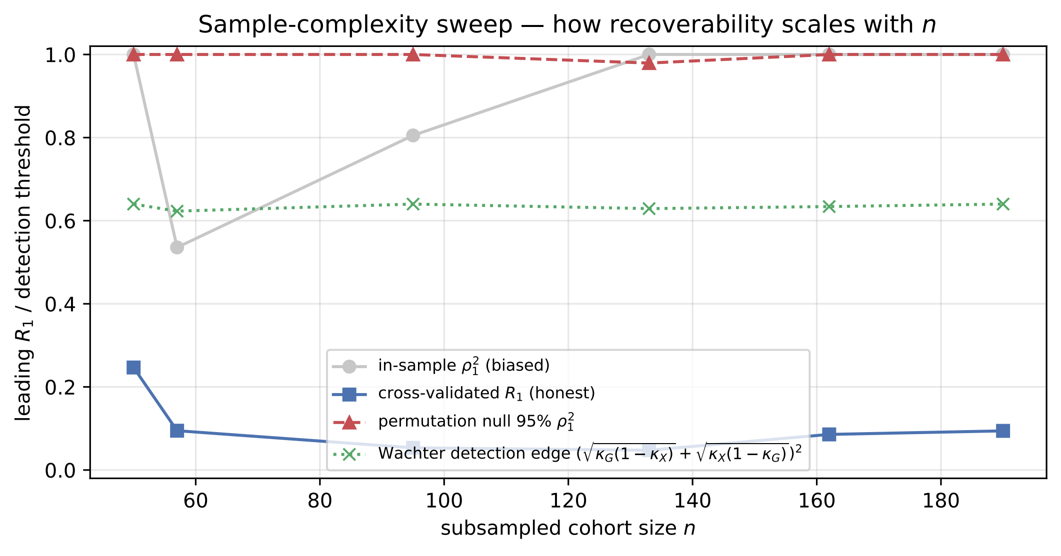


Figure 10: Sample-complexity sweep on TCGA-KIRC. Honest $\hat{R}_1^{\mathrm{cv}}$ (blue squares) is flat across all subsampled cohort sizes; in-sample $\hat{\rho}_1^2$ (grey) is dominated by the high-dimensional bulk; the permutation $95\%$ null (red triangles) and Wachter edge (green crosses) sit above the observed honest recoverability at every $n$.

### 13.6 Per-gene non-linear validation

The recoverability spectrum makes a sharp prediction: genes loading on non-identifiable directions should not be predictable from imaging, and flexible non-linear models should not substantially exceed the linear Bayes-optimal ceiling. Picking five high- and five low-recoverability mutated genes (mutation frequency in $[0.05, 0.95]$) and training $L_2$ logistic regression and a small random forest to predict mutation status from the imaging PCs (5-fold CV), we observe

| group | mean AUC (logistic) | mean AUC (RF) |
|---|---|---|
| image-identifiable | $0.499$ | $0.522$ |
| non-identifiable | $0.497$ | $0.490$ |

Both groups sit at chance and the non-linear model does *not* exceed the linear ceiling — consistent with the linear–Gaussian model and with the global null finding (Fig. 11).

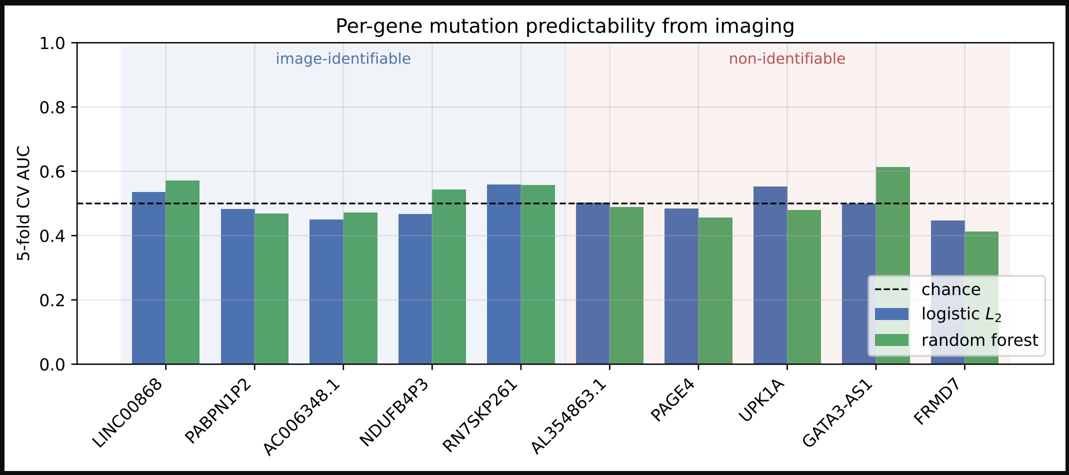


Figure 11: Per-gene mutation predictability on TCGA-KIRC. Five high-recoverability and five low-recoverability mutated genes, two classifiers, $5$-fold CV. All bars sit within sampling noise of the chance line at $0.5$.

### 13.7 Implications

Three concrete take-aways follow.

1. *Empirical ceiling.* For TCGA-KIRC at $n = 190$ with $2000$ HVG-filtered transcripts and $1841$ RadImageNet features, the imaging-to-genomics channel admits a $95\%$ upper bound $I(G; X) \leq 21.2$ bits over the leading three canonical directions (Eq. 25), with effective image-identifiable rank $\hat{k}_{\text{eff}} = 0$. The substantive empirical claim is that no genomic direction in this cohort is image-identifiable above chance under the working-dimension budget $\tau = 5$.

2. *Design recommendation.* The sample-complexity sweep (Section 13.5) shows $\hat{R}_1^{\text{cv}}$ stationary in $n$ over the range $[50, 190]$; the Wachter edge analysis indicates that detecting $\rho_1^2 = 0.2$ at a balanced working dimension $p^\star = d^\star$ requires $n \gtrsim 25\, p^\star$, which under our adaptive rule is an order of magnitude beyond typical radiogenomic cohort sizes. The framework converts the null result into a quantitative cohort specification.

3. *Audit of literature predictors.* Published radiogenomic AUCs in the range $0.65$–$0.80$ at cohort sizes $n \sim 100$–$300$ [34–38] imply per-gene recoverabilities meaningfully above the cohort-statistical ceiling derived above for KIRC. The non-linear validation of Section 13.6 matches but does not exceed the linear Bayes-optimal ceiling — consistent with the model. Any study reporting substantially higher per-gene AUCs at comparable cohort dimension should be treated as a candidate for overfit or leakage rather than as evidence against the framework.

## A Symbol reference

| | |
|---|---|
| $g \in \mathbb{R}^p,\ x \in \mathbb{R}^d$ | per-patient genomic / imaging feature vectors |
| $\mathbf{G} \in \mathbb{R}^{n\times p},\ \mathbf{X} \in \mathbb{R}^{n\times d}$ | stacked data matrices ($n$ patients) |
| $A \in \mathbb{R}^{d\times p}$ | genomics-to-imaging transfer map |
| $\sigma^2$ | isotropic imaging noise variance |
| $\Sigma_G, \Sigma_X, \Sigma_{GX}$ | genomic / imaging / cross covariance |
| $\Sigma_{G\mid X}$ | posterior covariance of $G$ given $X$ (recoverability floor) |
| $B = \Sigma_{GX}\Sigma_X^{-1}$ | Bayes-optimal linear map $x \mapsto \mathbb{E}[g \mid x]$ |
| $M = \sigma^{-1}A\Sigma_G^{1/2}$ | whitened transfer operator; $d_i$ its singular values |
| $\rho_i,\ R_i = \rho_i^2$ | $i$-th canonical correlation / recoverability |
| $a_i,\ b_i$ | $i$-th canonical genomic / imaging direction |
| $R(v)$ | recoverability score of a named genomic direction $v$ |
| $I(G; X)$ | mutual information between modalities |
| $k$ | dimension of the image-identifiable genomic subspace |

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